

Aryan Johari

Graduate Software Engineer

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Summary

Master's computing graduate (Unitec) who builds end-to-end software in Python and TypeScript, from ML research pipelines to privacy-aware APIs and WebGL front ends. Eligible to work in NZ (Post Study Work Visa; no sponsorship required). Seeking a graduate or junior software engineering role in Auckland; available from September 2026.

Skills

Languages: Python, TypeScript/JavaScript, SQL

ML / CV: PyTorch, computer vision, OpenCV, NLP (Presidio)

Web / Graphics: React, WebGL, Three.js, GLSL, Web Audio API

Backend: FastAPI, asyncio, REST APIs, Pydantic

Tools: Git, Docker, pytest, Vitest, Vite, Linux

Cloud / Data: AWS S3, PostgreSQL, Google Gemini API, Google Search Console

Education

Master of Applied Technology – Computing

Unitec Institute of Technology

Feb 2025 – June 2026

Auckland, New Zealand

- Thesis: video-level deepfake generator attribution (GSTF) using R(2+1)D, ArcFace metric learning, and continual learning under domain shift

Bachelor of Science in Computer Science

Thakur College of Science and Commerce

Sep 2021 – Apr 2024

Mumbai, India

- CGPA: 8.49

Projects

ADA

Python, asyncio, SQLite, Gemini API, AWS S3

Apr 2026 – May 2026

- Built a local asyncio AI agent harness with Gemini tool calling, SQLite persistence, and background workers for constrained edge hardware
- Designed deterministic ingest-to-publish workflows with quality gates and AWS S3 deployment; 531 pytest tests with GitHub Actions CI on Python 3.11/3.12

GSTF — Deepfake Video Attribution (Master's Thesis)

Python, PyTorch, Docker

Feb 2026 – May 2026

- Implemented Master's thesis PyTorch pipeline for video-level deepfake generator attribution using R(2+1)D, ArcFace metric learning, and EWC continual learning
- Achieved 86.67% held-out accuracy on FaceForensics++ with ArcFace; YAML-driven experiment runner with Grad-CAM attribution and Dockerised training

PII Gateway

Python, FastAPI, Presidio, Docker, PostgreSQL

Mar 2026 – May 2026

- Built Docker-first FastAPI gateway for PII redaction with Microsoft Presidio, policy-driven structured JSON field rules, and API-key authenticated sanitize endpoint
- Implemented batch ingest from PostgreSQL, CSV/JSON files, and S3-compatible storage with APScheduler jobs; 33 pytest tests and CI on Python 3.11/3.12

Background Studio

React, TypeScript, Three.js, WebGL, Vite

Apr 2026 – Jun 2026

- Built browser-based WebGL studio for designing animated hero backgrounds with real-time GPU compositing and versioned preset JSON export for site embeds
- Authored single-pass GLSL fragment shader with per-layer effect bundles; 82 Vitest tests for preset validation and optional Vercel serverless mood API

Sound Visualiser

Mar 2026 – Jul 2026

TypeScript, Three.js, Web Audio API, GLSL

- Built real-time audio-reactive WebGL visualiser mapping Meyda spectral features to cinematic 3D, live webcam, and acid feedback modes
- Implemented beat-sync PLL, ping-pong FBO temporal feedback, and frame-rate-independent audio smoothing across three visual modes

Experience

SEO Specialist

Specialist Support Services

*Jul 2025 – Present
Auckland, New Zealand*

- Scaled monthly organic impressions 414% (1,080 to 5,555) over 11-month GSC window; 2,595 clicks, 6.81% CTR, and 8.31 average position across three sites
- Engineered GSC infrastructure, indexing workflows, mobile-first layouts (1,608 mobile clicks), and pSEO page templates; position 1.63 for “em bail accommodation”

Junior Website Developer

KRIL Digital

*May 2024 – Oct 2024
Mumbai, India*

- Built and maintained approximately 20 client websites (WordPress, React, Shopify) across industrial, education, and events sectors
- Improved average site performance via daily updates and SEO best practices: 70% faster load times, 45% visibility uplift, and 30% higher CTR